

Supplementary Information for Scaling of Berry-curvature monopole dominated large linear positive magnetoresistance

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Supplementary Information Text

1. Single crystal growth

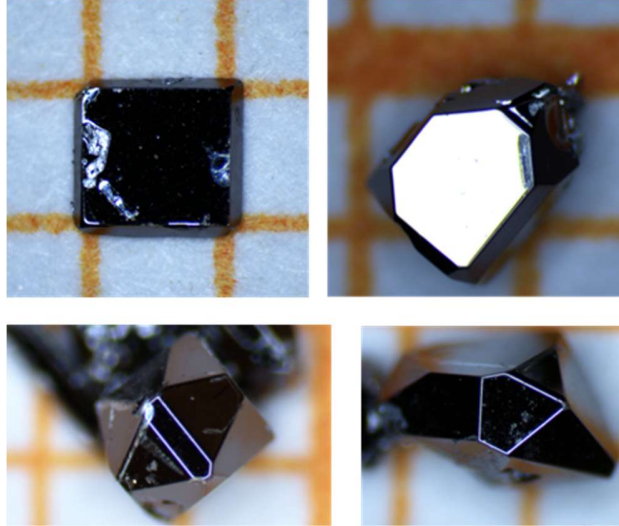


Fig. S1 | Optical images of single crystals grown by a flux method

The single crystals of CoS_2 were grown by a flux method in a Al_2O_3 crucible sealed in a quartz tube. The quartz tube was heated to 800°C in 12 hours and kept there for 24 hours, to make sulphur and melting cobalt react sufficiently, then slowly cooled to 600°C in 7 days for CoS_2 single crystals growth. Then the quartz tube was transferred from the muffle furnace at 600°C to a centrifugal machine where the flux and CoS_2 single crystals were separated.

The chemical composition of single crystals was confirmed by Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP-AES). The result showed a slight cobalt deficiency in our CoS_2 single crystals.

Table S1 | Chemical composition of CoS_2 single crystals

Co	S	Co : S
32.79	67.21	0.98 : 2

2. Crystal structure and X-ray diffraction

The single crystals generally showed a cubic shape which is consistent with the unit cell (Fig. 1a). In the unit cell, the Co atoms form a face-centered cube, but the positions of S atoms are difficult to describe. Focusing on the basis, the Co and S atoms form an octahedron where the Co atom locate at the center. Setting the position of Co atom as origin, six S positions can be expressed as $(\frac{1}{2}-\alpha, -\alpha, \alpha)$, $(\alpha, \frac{1}{2}-\alpha, \alpha)$, $(-\alpha, -\alpha, \frac{1}{2}-\alpha)$ and their opposite coordinates (Fig. S2a).

Here, α is reduced length, satisfying $0 \leq \alpha < 1/2$, as 1 represents the lattice parameter a . Therefore, the

distances between Co atom and six S atoms are all equal to $a\sqrt{2\alpha^2 + \left(\frac{1}{2}-\alpha\right)^2}$. Here, a is the lattice parameter. The Co-S octahedron is not a regular octahedron, as the Co-S bonds are not perpendicular to each other.

After refinement of the powder X-ray diffraction spectra at different temperatures from 5 to 300 K, we extracted two pieces of information on the CoS_2 unit cell, lattice parameter and S atom position α (Fig. S2b). When the temperature reduces to approximately 25 K, an anomalous change can be

observed in both the lattice parameter and the position of the S atom. The lattice parameter increases and S atom position α decreases, which means that the CoS₂ unit cell expands and the Co-S bonds rotate when cooling down to 25 K.

Although the lattice parameter a and S coordinate α change with temperature, it is difficult to describe how the Co-S octahedron change with temperature. Fortunately, the projection of Co-S octahedron on the xy plane is definitely a square, then we can imagine the change of the 3D Co-S octahedron by studying the change of the 2D projection square. The projection of Co-S octahedron on the xy plane is a quadrangle constituted by four S atoms (Fig. S2a), whose coordinates are $(\frac{1}{2}-\alpha, -\alpha)$, $(\alpha, \frac{1}{2}-\alpha)$, $(\alpha-\frac{1}{2}, \alpha)$, $(-\alpha, \alpha-\frac{1}{2})$. It is easy to prove that the projection quadrangle is a square. We can describe the change of the Co-S square by the projected length of Co-S bond $l = a\sqrt{\alpha^2 + (\frac{1}{2}-\alpha)^2}$, which decides the scaling of the Co-S square, and the angle between the projected Co-S bond and positive x -axis $\theta = \arctan \frac{\alpha}{1/2-\alpha}$, which decides the rotation of the Co-S square.

We determined that the Co-S bond rotates counter-clockwise and becomes longer when the temperature drops to 25 K (Fig. S2c). This negative thermal expansion is similar to the behavior observed in ZrW₂O₈, where the rotation of the WO₃ octahedron results in a notable negative expansion(1, 2).

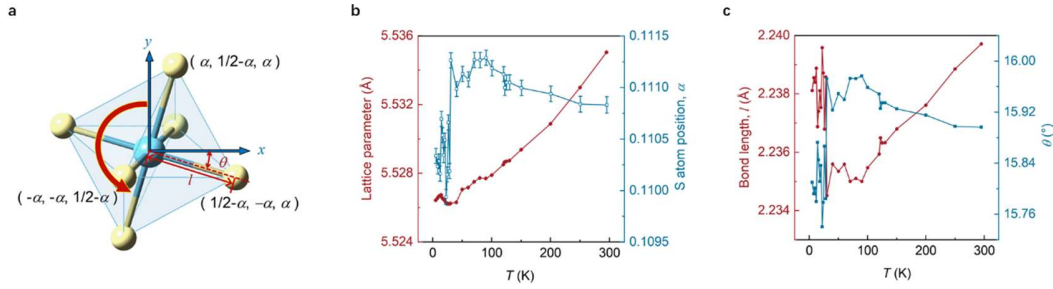


Fig. S2 |Rotation of the Co-S octahedron and the negative thermal expansion. a, Co-S octahedron. Co atom is located at the ordinate origin, and S atom position is represented by $(1/2 - \alpha, -\alpha, \alpha)$. l is the projected length of the Co-S bond on xy plane, and θ is the angle between the projection of the Co-S bond and positive x -axis. The Co-S bond and the octahedron rotates counter-clockwise when cooling down to 25 K. **b**, Lattice parameter and S atom position α extracted from refinement of powder X-ray diffraction spectrums. The lattice parameter increases and S atom position α decreases anomalously when temperature decreases to approximately 25 K. **c**, Temperature dependence of l and θ in **a**.

3. Weak magnetic anisotropy

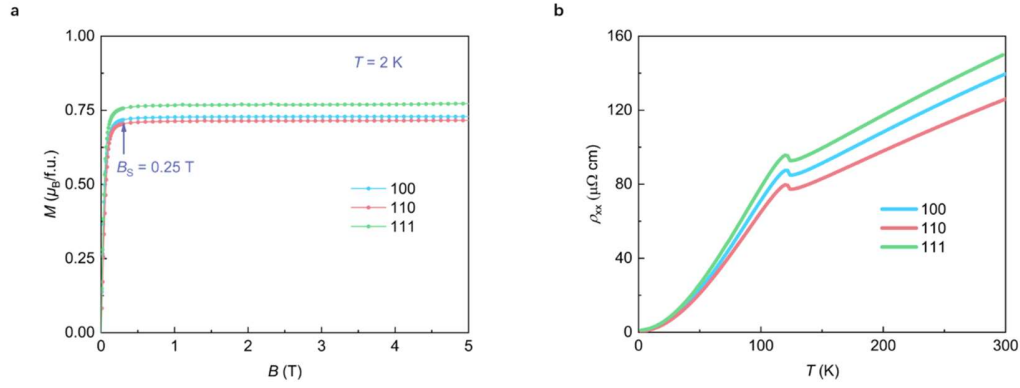


Fig. S3 | Magnetic and electrical transport measurements. **a**, Field dependence of magnetization at 2 K with the magnetic field along [100], [110], [111]. **b**, Temperature dependence of longitudinal resistivity with the magnetic field along [100], [110], [111].

We measured field-dependent magnetization and temperature-dependent longitudinal resistivity of three samples with the magnetic field along [100], [110], [111] respectively. Both magnetic and electrical measurements show slight difference when the magnetic field along different directions.

4. Theoretical calculations

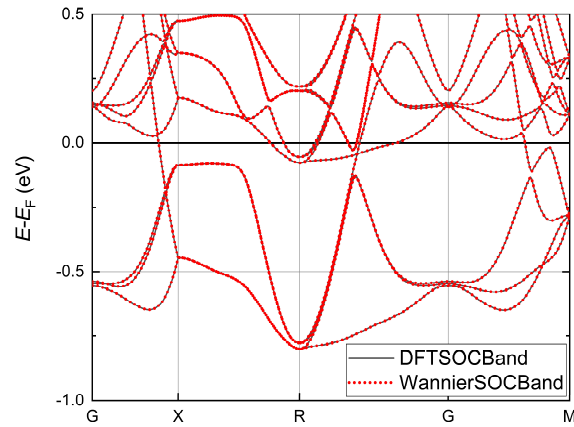


Fig. S4 | Band structure of CoS_2 with SOC included. Solid lines: band structure from a first-principles DFT calculation. Red-dotted lines: Wannier-interpolated band structure. Wannier-interpolated band structure coincide well with DFT-calculated band structure.

Table S2 | Positions and energies of eight pairs of Weyl nodes

k_1	k_2	k_3	Chirality	Energy (meV)
0.00007	-0.44121	0.40712	1	10.52259
0.00021	0.44121	-0.40712	-1	10.52192
0.00016	0.44118	0.40716	1	10.56674
0.00031	-0.44118	-0.40716	-1	10.56661
0.40613	0.00025	-0.44224	1	10.10003
-0.40613	0.00022	0.44224	-1	10.09980
-0.40617	-0.00042	-0.44221	1	10.05823
0.40617	-0.00041	0.44221	-1	10.05880
-0.15729	0.40783	-0.32540	1	-1.57742
0.15520	-0.40824	0.32657	-1	-1.39112
-0.15517	-0.40828	-0.32654	1	-1.35757
0.15719	0.40787	0.32541	-1	-1.53744
0.15603	-0.40809	-0.32614	1	-1.49953
-0.15627	0.40803	0.32601	-1	-1.52628
0.15637	0.40802	-0.32592	1	-1.48732
-0.15587	-0.40813	0.32620	-1	-1.43998

The positions of Weyl nodes in Brillouin zone were calculated. Within the range of 10 meV around Fermi level, 8 pairs of Weyl nodes exist. As shown in Fig. 2b and Table S2, the Weyl nodes gather around M points in the Brillouin zone and their positions are close to the Fermi surface. In wider energy (± 30 meV) ranges, we observed more linear-dispersion band crossings (Fig. S5).

Based on the measured AHC and cobalt deficiency, we inferred that the Fermi energy in real materials is about 20 meV lower than calculation (Fig. 2c). The offset of Fermi energy is so small that the calculated Weyl nodes can still affect the transport behaviors.

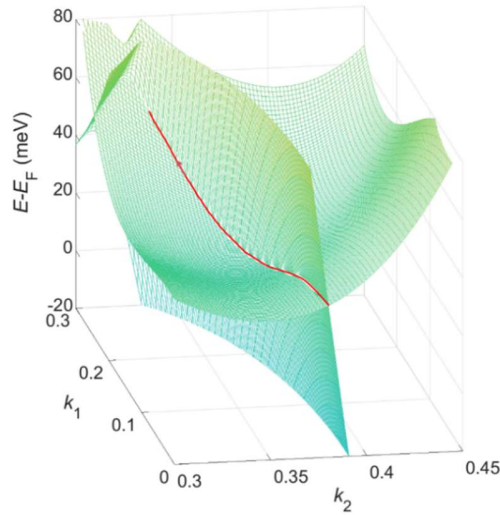


Fig. S5 | Linear-dispersion band crossings around the Fermi energy in k_x - k_y plane ($k_z = 0$).

5. TYJ fitting

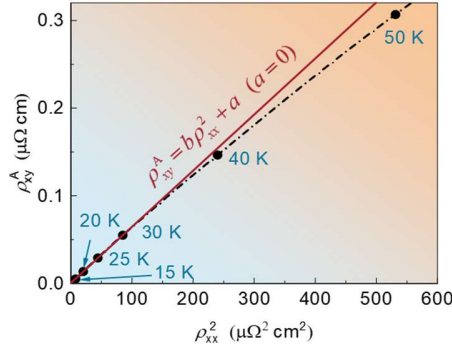


Fig. S6 | The anomalous Hall resistivity versus the square of the longitudinal resistivity from 15 to 50 K.

We separated the intrinsic and extrinsic anomalous Hall components from the measured AHR using the TYJ model(3). Clearly, at least below 30 K, the anomalous Hall effect in CoS₂ is completely dominated by the intrinsic mechanism, Berry curvature, because the AHR is proportional to the square of the longitudinal resistivity.

6. Low-field and high-field behaviors

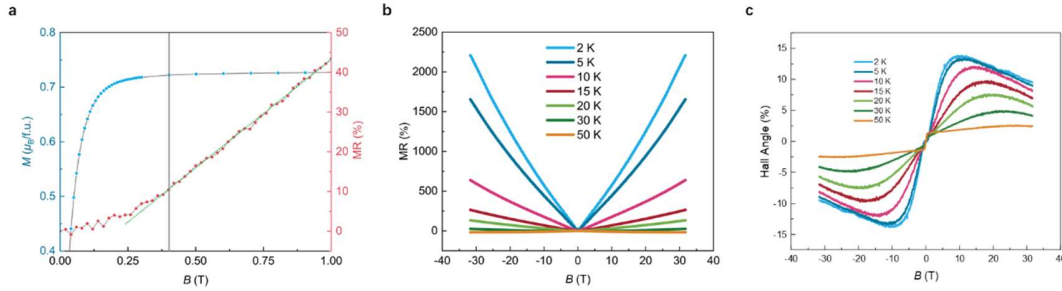


Fig. S7 | Non-linear behavior in MR, and the high-field Hall angle. a, Low-field magnetization and MR. **b,** Field dependence of MR with the magnetic field up to 32 T. **c,** Field dependence of the Hall angle with the magnetic field up to 32 T.

The MR below 0.4 T shows a near field-square behavior, and the magnetization shows an unsaturation behavior at the same magnetic field region. Therefore, at low fields, the MR behavior may be affected by the magnetic scattering.

When the magnetic field increases to 32 T, the positive MR does not show any signature of saturation. And the MR shows a slight deviation from the linear relation. The high-field behavior is not the same in different systems(4-8). The deviation may result from the influence of the magnetic field on the Berry curvature, because it is impossible that the Berry curvature remains unchanged when strong magnetic field changes the energy band significantly.

The maximum of the Hall angle is measured at 2 K, 10 T. The value is less than 0.15, which means the guiding center theory(9, 10) is not satisfied in CoS₂.

7. Berry curvature induced AHE and linear positive MR

In our model, the influence of the Berry curvature on MR originates from the influence of the Berry curvature on electron motion. The Berry-curvature-modified motion equations of Bloch electrons(11) are given by

$$\dot{\mathbf{r}} = \frac{1}{\hbar} \nabla_{\mathbf{k}} \varepsilon - \dot{\mathbf{k}} \times \boldsymbol{\Omega} \quad (1)$$

$$\hbar \dot{\mathbf{k}} = -e\mathbf{E} - e\dot{\mathbf{r}} \times \mathbf{B} \quad (2)$$

We assume that $\mathbf{E} \perp \mathbf{B}$, which is consistent with the electrical transport measurement condition.

Suppose $\mathbf{v}_k = \frac{1}{\hbar} \nabla_{\mathbf{k}} \varepsilon$, then the motion equations can be rewritten as

$$\dot{\mathbf{r}} = \mathbf{v}_k + \frac{e}{\hbar} \mathbf{E} \times \boldsymbol{\Omega} + \frac{e}{\hbar} (\dot{\mathbf{r}} \times \mathbf{B}) \times \boldsymbol{\Omega} \quad (3)$$

$$\dot{\mathbf{k}} = -\frac{e}{\hbar} \left[\mathbf{E} + \mathbf{v}_k \times \mathbf{B} - (\dot{\mathbf{k}} \times \boldsymbol{\Omega}) \times \mathbf{B} \right] \quad (4)$$

In Eq. (3), the first term is the conventional velocity, the second term is the intrinsic Hall velocity, and the last term is an anomalous velocity about the Berry curvature and magnetic field. The anomalous velocity is in the opposite direction to the conventional velocity when the Berry curvature is along the magnetic field. In Eq. (4), the first term is the electrical field force, the second term is the Lorentz force, and the last term is an anomalous force in the opposite direction to the electrical field force and the Lorentz force.

Suppose $D(\mathbf{k}) = 1 + \frac{e}{\hbar} B \Omega_z(\mathbf{k})$ to represent the modified density of states. The motion equations are transformed to

$$\dot{\mathbf{r}} = \frac{1}{D(\mathbf{k})} \left[\mathbf{v}_k + \frac{e}{\hbar} \mathbf{E} \times \boldsymbol{\Omega} + \frac{e}{\hbar} (\mathbf{v}_k \cdot \boldsymbol{\Omega}) \mathbf{B} \right] \quad (5)$$

$$\dot{\mathbf{k}} = -\frac{e}{\hbar} \frac{1}{D(\mathbf{k})} (\mathbf{E} + \mathbf{v}_k \times \mathbf{B}) \quad (6)$$

Because the electrical signal parallel to the magnetic field is not interested in this study, we can ignore the last term in Eq. (5). The final form of motion equations we need becomes

$$\dot{\mathbf{r}} = \frac{1}{D(\mathbf{k})} \left(\mathbf{v}_k + \frac{e}{\hbar} \mathbf{E} \times \boldsymbol{\Omega} \right) \quad (7)$$

$$\dot{\mathbf{k}} = -\frac{e}{\hbar} \frac{1}{D(\mathbf{k})} (\mathbf{E} + \mathbf{v}_k \times \mathbf{B}) \quad (8)$$

The motion equations can describe an individual electron. When we study the electrical transport properties in real materials, we should integrate the velocity of all electrons in the k -space to get the current density. The current density expression with modified density of states is

$$\mathbf{j} = -e \int \frac{d\mathbf{k}}{(2\pi)^3} D(\mathbf{k}) \dot{\mathbf{r}} f \quad (9)$$

Here, f represents the distribution function.

The distribution function is solved from the Boltzmann transport equation with a relaxation-time approximation.

$$\dot{\mathbf{r}} \cdot \nabla_{\mathbf{r}} f + \dot{\mathbf{k}} \cdot \nabla_{\mathbf{k}} f = \frac{f_0 - f}{\tau} \quad (10)$$

Here, $f_0 = \frac{1}{1 + e^{(\varepsilon - \varepsilon_F)/k_B T}}$ represents the Fermi-Dirac distribution function. More precisely, the Boltzmann transport equation is

$$\dot{\mathbf{r}} \cdot \frac{\partial f}{\partial T} \nabla_{\mathbf{r}} T + \dot{\mathbf{k}} \cdot \frac{\partial f}{\partial \varepsilon} \nabla_{\mathbf{k}} \varepsilon = \frac{f_0 - f}{\tau} \quad (11)$$

With no temperature gradient on materials, the first term in Eq. (11) should reduce to zero. Then we plug Eq. (8) into Eq. (11), and approximate that $\frac{\partial f}{\partial \varepsilon} \approx \frac{\partial f_0}{\partial \varepsilon}$. Then we get the distribution function

$$f = f_0 + \frac{e\tau}{D(\mathbf{k})} \frac{\partial f_0}{\partial \varepsilon} (\mathbf{E} \cdot \mathbf{v}_k) \quad (12)$$

Plug Eq. (7) and Eq. (12) into Eq. (9), and ignore the square term of electrical field, then the current density becomes

$$\mathbf{j} = -e \int \frac{d\mathbf{k}}{(2\pi)^3} \left[\mathbf{v}_k f_0 + \frac{e\tau}{D(\mathbf{k})} \frac{\partial f_0}{\partial \varepsilon} (\mathbf{E} \cdot \mathbf{v}_k) \mathbf{v}_k + \frac{e}{\hbar} \mathbf{E} \times \boldsymbol{\Omega} f_0 \right] \quad (13)$$

The first term is the current in equilibrium state, which should be zero. The second term is the current induced by the electrical field. The third term is the intrinsic anomalous Hall current. Therefore, the longitudinal conductivity and normal Hall conductivity can be derived from the second term, and the anomalous Hall conductivity can be derived from the third term.

$$\sigma_{xx} = -e^2 \tau \int \frac{d\mathbf{k}}{(2\pi)^3} \frac{\partial f_0}{\partial \varepsilon} \frac{1}{D(\mathbf{k})} v_x^2 \quad (14)$$

$$\sigma_{yx}^N = e^2 \tau \int \frac{d\mathbf{k}}{(2\pi)^3} \frac{\partial f_0}{\partial \varepsilon} \frac{1}{D(\mathbf{k})} v_x v_y \quad (15)$$

$$\sigma_{yx}^A = \frac{e^2}{\hbar} \int \frac{d\mathbf{k}}{(2\pi)^3} \Omega_z f_0 \quad (16)$$

Generally, the energy dispersion is an even function, so v_x and v_y are odd functions. Thus, the normal Hall conductivity (Eq.(15)) is zero.

To get the concrete equations of conductivity, we consider the simplest 3D Weyl model. A pair of Weyl nodes are located at $(0, 0, \pm c)$ in the k -space. The Hamiltonian of the Weyl node at $(0, 0, c)$ is

$$H = \lambda[k_x \sigma_x + k_y \sigma_y - (k_z - c) \sigma_z] + \Delta \quad (17)$$

Thus, the dispersions of energy bands are

$$\varepsilon_{\pm} = \Delta \pm \lambda \sqrt{k_x^2 + k_y^2 + (k_z - c)^2} \quad (18)$$

Δ represents the shift from the Fermi energy, and λ represents the dispersion slope.

The Berry curvature created by the Weyl node is

$$\boldsymbol{\Omega}_{\pm}(\mathbf{k}) = \pm \frac{1}{2\sqrt{k_x^2 + k_y^2 + (k_z - c)^2}} [k_x, k_y, k_z - c] \quad (19)$$

According to space-inversion symmetry, we can easily get the dispersions and the Berry curvature of the Weyl node at $(0, 0, -c)$. The z -direction Berry curvature created by a pair of Weyl nodes is symmetrical.

When the Weyl nodes are in a magnetic field or a molecular field of a magnetic material, only the band structures between $k_z = \pm c$ are occupied according to the lowest Landau band(12). This means that we only need to consider the band structures and Berry curvature between $k_z = \pm c$ when integrating.

Now, assume $\Delta > 0$, then consider a simple situation, 0 K. The AHC can be calculated as

$$\sigma_{yx}^A = 2 \frac{e^2}{\hbar} \int \frac{d\mathbf{k}}{(2\pi)^3} \Omega_z^- \Theta(-\varepsilon_-) \quad (20)$$

where $\Theta(x)$ is the step function, and the parameter 2 comes from the symmetrical Berry curvature of two Weyl nodes. Turning rectangular coordinates into cylindrical coordinates, continue the integral

$$\sigma_{yx}^A = -\frac{e^2}{4\pi^2\hbar} \left(\int_{-c}^{c-\frac{\Delta}{\lambda}} dk_z \int_0^\infty d\rho + \int_{c-\frac{\Delta}{\lambda}}^c dk_z \int_{\sqrt{\frac{\Delta^2}{\lambda^2} - (k_z - c)^2}}^\infty d\rho \right) \frac{\rho(k_z - c)}{\sqrt{\rho^2 + (k_z - c)^2}^3} = \frac{e^2}{4\pi^2\hbar} \left(2c - \frac{\Delta}{2\lambda} \right) \quad (21)$$

And we can find that the AHC is positively associated with the distance between two Weyl nodes, and is negatively associated with the shift from the Fermi energy. This conclusion is consistent with the known theory about the intrinsic anomalous Hall effect. Assuming that $c \gg \frac{\Delta}{\lambda}$, the AHC is $\frac{e^2}{ha} \sim 10^3 \Omega^{-1}\text{cm}^{-1}$, where a is the lattice parameter.

The longitudinal conductivity can be calculated as

$$\sigma_{xx} = \frac{2e^2\tau}{\hbar^2} \int \frac{d\mathbf{k}}{(2\pi)^3} \frac{(\partial_{k_x} \varepsilon_-)^2}{1 + \frac{e}{\hbar} B \Omega_z^-} \delta(\varepsilon_-) \quad (22)$$

where $\delta(x)$ is the delta function. In Eq. (22), the term contains magnetic field is in the denominator, and the integral would give the relation $\sigma_{xx} - \sigma_{xx}(B=0) \propto \frac{1}{B}$, which is equal to linear magnetoresistance.

For a concise result, we make the approximations that

$$\Omega_F = \frac{\int \frac{d\mathbf{k}}{(2\pi)^3} \Omega_z^- \delta(\varepsilon_-)}{\int \frac{d\mathbf{k}}{(2\pi)^3} \delta(\varepsilon_-)} \quad (23)$$

$$\sigma_{xx} = \frac{2e^2\tau}{\hbar^2 (1 + \frac{e}{\hbar} B \Omega_F)} \int \frac{d\mathbf{k}}{(2\pi)^3} (\partial_{k_x} \varepsilon_-)^2 \delta(\varepsilon_-) \quad (24)$$

According to Eq. (24), the magnetoresistance (MR) is

$$\text{MR} = \frac{\rho_{xx}(B) - \rho_{xx}(B=0)}{\rho_{xx}(B=0)} = \frac{\sigma_{xx}(B=0)}{\sigma_{xx}(B)} - 1 = \frac{e}{\hbar} B \Omega_F \quad (25)$$

Here, we assume that the Hall conductivity is much less than the longitudinal conductivity, which is reasonable for the small Hall angle in CoS₂ (Fig. S7c). Thus, we achieve the conclusion that the linear positive magnetoresistance originates from the average of the Berry curvature on the Fermi surface. And turning rectangular coordinates into cylindrical coordinates, calculate the MR at 0 K as

$$\text{MR} = \frac{e}{\hbar} B \frac{\int_0^\infty dk \int_{\frac{\pi}{2}}^\pi d\theta k^2 \sin\theta \frac{-\cos\theta}{2k^2} \delta(k - \frac{\Delta}{\lambda})}{\int_0^\infty dk \int_{\frac{\pi}{2}}^\pi d\theta k^2 \sin\theta \delta(k - \frac{\Delta}{\lambda})} = \frac{e\lambda^2}{4\hbar\Delta^2} B \quad (26)$$

And we find that the slope of LPMR is positively associated with the dispersion slope, and is negatively associated with the shift from the Fermi energy. Assuming that $\Delta \sim 1$ meV and $\lambda \sim 1$ eV Å, the MR at 10 T is $\frac{e\lambda^2}{4\hbar\Delta^2} B \sim 1000$ %.

Here we provide one proof method connecting theory and experimental data that is to fit the temperature-dependent AHC and slopes of MR curves with the temperature-dependent Berry curvature integral.

Consider a finite temperature and assume that $\lambda c / k_B T \gg 1$, then the AHC can be calculated as

$$\begin{aligned} \sigma_{yx}^A &= \frac{e^2}{4\pi^2 \hbar} \int_{-2c}^0 dk_z \int_0^\infty d\rho \frac{-\rho k_z}{\sqrt{\rho^2 + k_z^2}^3} \left(\frac{1}{1 + e^{\varepsilon_- / k_B T}} - \frac{1}{1 + e^{\varepsilon_+ / k_B T}} \right) \\ &\approx \frac{e^2}{4\pi^2 \hbar} \left[\left(\int_{-2c}^{-\frac{\Delta}{\lambda}} dk_z \int_0^\infty d\rho + \int_{-\frac{\Delta}{\lambda}}^0 dk_z \int_{\sqrt{\frac{\Delta^2}{\lambda^2} - k_z^2}}^\infty d\rho \right) \frac{-\rho k_z}{\sqrt{\rho^2 + k_z^2}^3} (1 - \frac{e^{-\varepsilon_- / 2k_B T}}{2}) \right. \\ &\quad \left. + \int_{-\frac{\Delta}{\lambda}}^0 dk_z \int_0^{\sqrt{\frac{\Delta^2}{\lambda^2} - k_z^2}} d\rho \frac{-\rho k_z}{\sqrt{\rho^2 + k_z^2}^3} \frac{e^{-\varepsilon_- / 2k_B T}}{2} + \int_{-2c}^0 dk_z \int_0^\infty d\rho \frac{\rho k_z}{\sqrt{\rho^2 + k_z^2}^3} \frac{e^{-\varepsilon_+ / 2k_B T}}{2} \right] \\ &\approx \frac{e^2}{4\pi^2 \hbar} (2c - \frac{\Delta}{2\lambda} - \frac{k_B T}{\lambda} e^{-\frac{\Delta}{2k_B T}}) \end{aligned} \quad (27)$$

And the average of the Berry curvature on the Fermi surface can be calculated as

$$\Omega_F = \frac{\int \frac{d\mathbf{k}}{(2\pi)^3} \Omega_z \frac{\partial f_0}{\partial \varepsilon}}{\int \frac{d\mathbf{k}}{(2\pi)^3} \frac{\partial f_0}{\partial \varepsilon}} = \frac{1}{4} \frac{\int_0^\infty dk \left[\frac{e^{\frac{\Delta - \lambda k}{k_B T}}}{(1 + e^{\frac{\Delta - \lambda k}{k_B T}})^2} - \frac{e^{\frac{\Delta + \lambda k}{k_B T}}}{(1 + e^{\frac{\Delta + \lambda k}{k_B T}})^2} \right]}{\int_0^\infty dk k^2 \left[\frac{e^{\frac{\Delta - \lambda k}{k_B T}}}{(1 + e^{\frac{\Delta - \lambda k}{k_B T}})^2} + \frac{e^{\frac{\Delta + \lambda k}{k_B T}}}{(1 + e^{\frac{\Delta + \lambda k}{k_B T}})^2} \right]} = \frac{\lambda^2}{4\Delta^2} \frac{\tanh(\frac{\Delta}{2k_B T})}{1 + \frac{\pi^2 k_B^2 T^2}{3\Delta^2}} \quad (28)$$

Then the slope of MR is

$$\frac{\text{MR}}{B} = \frac{e}{\hbar} \Omega_F = \frac{e\lambda^2}{4\hbar\Delta^2} \frac{\tanh(\frac{\Delta}{2k_B T})}{1 + \frac{\pi^2 k_B^2 T^2}{3\Delta^2}} \quad (29)$$

Combined with the results assuming $\Delta < 0$, the temperature-dependent AHC and slopes of MR are

$$\sigma_{yx}^A(T) = \frac{e^2}{4\pi^2 \hbar} (2c - \frac{|\Delta|}{2\lambda} - \frac{k_B T}{\lambda} e^{-\frac{|\Delta|}{2k_B T}}) \quad (30)$$

$$\frac{\text{MR}}{B}(T) = \frac{e\lambda^2}{4\hbar\Delta^2} \frac{\tanh(\frac{\Delta}{2k_B T})}{1 + \frac{\pi^2 k_B^2 T^2}{3\Delta^2}} \quad (31)$$

The Eq. (31) tells us that the sign of MR is the same as the sign of Δ , which means that linear negative MR can also be generated. However, the reported topological materials did not show linear negative MR phenomenon. The reason may be that negative MR should not be larger than 100%, so that the negative MR could be ignored in report or be covered by other mechanisms, two-carrier model for example. More research is need in the future.

In summary, the intrinsic model can well describe the trend of AHC and LPMR changing with temperature as shown in Fig. 4b and 4c. The fitting parameters are shown in Table S3. The fitting parameters from AHC and LPMR are in different range. For example, Δ of CoS₂ differs by one order of magnitude.

We also consider multiple-Weyl model, and find there is always some mismatch in fitting parameters. We consider n pairs of Weyl nodes with different parameters for each pair, which are denoted as Δ_i, λ_i, c_i . Then the longitudinal and anomalous Hall conductivity can be described as

$$\sigma_{xx} = \sum_{i=1}^n \frac{\sigma_i}{1 + \frac{e}{\hbar} B\Omega_i} \quad (32)$$

$$\sigma_{yx}^A(T) = \frac{e^2}{4\pi^2\hbar} \sum_{i=1}^n (2c_i - \frac{|\Delta_i|}{2\lambda_i} - \frac{k_B T}{\lambda_i} e^{-\frac{|\Delta_i|}{2k_B T}}) \quad (33)$$

where $\sigma_i = \frac{2e^2\tau_i}{\hbar^2} \int \frac{d\mathbf{k}}{(2\pi)^3} (\partial_{k_x} \varepsilon_i)^2 \frac{\partial f_0}{\partial \varepsilon_i}$ and $\Omega_i = \frac{\lambda_i^2}{4\Delta_i^2} \frac{\tanh(\frac{\Delta_i}{2k_B T})}{1 + \frac{\pi^2 k_B^2 T^2}{3\Delta_i^2}}$. Then the MR is

$$\text{MR} = \frac{\sigma_{xx}(B=0)}{\sigma_{xx}(B)} - 1 = \frac{\sum_{i=1}^n \sigma_i}{\sum_{i=1}^n \frac{\sigma_i}{1 + \frac{e}{\hbar} B\Omega_i}} - 1 \quad (34)$$

The slope of MR is

$$\frac{\partial \text{MR}}{\partial B} = \frac{\sum_{i=1}^n \sigma_i \sum_{i=1}^n \frac{\frac{e}{\hbar} \sigma_i \Omega_i}{(1 + \frac{e}{\hbar} B\Omega_i)^2}}{(\sum_{i=1}^n \frac{\sigma_i}{1 + \frac{e}{\hbar} B\Omega_i})^2} \xrightarrow{\frac{e}{\hbar} B\Omega_i \gg 1} \frac{e}{\hbar} \frac{\sum_{i=1}^n \sigma_i}{\sum_{i=1}^n \frac{\sigma_i}{\Omega_i}} \quad (35)$$

From Eqs.(33)(35), we find that Δ_i mainly affects the changing with temperature, λ_i and c_i affect the values of AHC and LPMR. For multiple Weyl models with different Δ_i , the curves of temperature-dependent AHC and LPMR would not be as simple as those of single Weyl model. Furthermore, this multiple Weyl model will result in the overparametrization during the experimental fitting.

In order to reduce the parameters, we consider n pairs of Weyl nodes with the same Δ, λ, c . Then the temperature-dependent longitudinal and anomalous Hall conductivity can be described as

$$\sigma_{xx} = -\frac{ne^2\tau}{1 + \frac{e}{\hbar}B\Omega_F} \int \frac{d\mathbf{k}}{(2\pi)^3} \frac{\partial f_0}{\partial \varepsilon} v_x^2 \quad (36)$$

$$\sigma_{yx}^A(T) = \frac{ne^2}{4\pi^2\hbar} \left(2c - \frac{|\Delta|}{2\lambda} - \frac{k_B T}{\lambda} e^{-\frac{|\Delta|}{2k_B T}} \right) \quad (37)$$

According to Eq.(36), the slope of MR is

$$\frac{\text{MR}}{B}(T) = \frac{e}{\hbar} \Omega_F(T) = \frac{e\lambda^2}{4\hbar\Delta^2} \frac{\tanh(\frac{\Delta}{2k_B T})}{1 + \frac{\pi^2 k_B^2 T^2}{3\Delta^2}}, \quad (38)$$

and is just the same as the Eq.(31). After fitting with experimental AHC, c would shrink n times and λ would expand n times than those of single Weyl model. The fitting parameters of multiple Weyl model seems more reasonable. However, it is still difficult to explain both AHC and LPMR by using the same set of parameters. Thus, the parameter mismatch mainly results from other reasons.

The first reason is that we made a Berry curvature approximation in longitudinal conductivity (Eqs.(22)-(24)), where we replaced the Berry curvature Ω_z with the average value of Berry curvature Ω_F . The second reason is that the band structures in real material are more complex. We ignored the energy gap, band tilt, and arbitrary coordinates to avoid overparameterization in fitting. The third reason is that Berry curvature is provided by not only Weyl nodes but also other band structures in real materials. These could result in the inaccuracy of fitting parameters.

Table S3 | Fitting parameters of the intrinsic model

	Materials	Δ (meV)	λ (eV Å)	$2c$ (Å⁻¹)
AHC	CoS ₂	10.65	0.0078	1.79
	Co ₃ Sn ₂ S ₂	80.19	0.0008	49.45
	Co ₂ MnGa	11.82	0.0131	2.93
LPMR	CoS ₂	1.26	0.5300	
	MnBi	4.09	1.1981	
	PrAlSi	3.10	0.7282	
	FeP	3.81	2.2020	
	NbP	8.26	148.8389	
	ZrSiS	7.53	61.1726	

SI References

1. Mary T A, Evans J S O, Vogt T, Sleight A W (1996) Negative Thermal Expansion from 0.3 to 1050 Kelvin in ZrW_2O_8 . *Science* 272: 90-92.
2. Ernst G, Broholm C, Kowach G R, Ramirez A P (1998) Phonon density of states and negative thermal expansion in ZrW_2O_8 . *Nature* 396: 147-149.
3. Tian Y, Ye L, Jin X (2009) Proper scaling of the anomalous Hall effect. *Phys Rev Lett* 103: 087206.
4. Novak M, Sasaki S, Segawa K, Ando Y (2015) Large linear magnetoresistance in the Dirac semimetal TlBiSSe . *Phys Rev B* 91: 041203.
5. Hu J et al. (2016) π Berry phase and Zeeman splitting of Weyl semimetal TaP . *Sci Rep* 6: 18674.
6. He Y et al. (2021) Large linear non-saturating magnetoresistance and high mobility in ferromagnetic MnBi . *Nat Commun* 12: 4576.
7. Lyu M et al. (2020) Nonsaturating magnetoresistance, anomalous Hall effect, and magnetic quantum oscillations in the ferromagnetic semimetal PrAlSi . *Phys Rev B* 102: 085143.
8. Takiguchi K et al. (2020) Quantum transport evidence of Weyl fermions in an epitaxial ferromagnetic oxide. *Nat Commun* 11: 4969.
9. Song J C W, Refael G, Lee P A (2015) Linear magnetoresistance in metals: Guiding center diffusion in a smooth random potential. *Phys Rev B* 92.
10. Leahy I A et al. (2018) Nonsaturating large magnetoresistance in semimetals. *Proc Natl Acad Sci U S A* 115: 10570-10575.
11. Xiao D, Chang M-C, Niu Q (2010) Berry phase effects on electronic properties. *Rev Mod Phys* 82: 1959-2007.
12. Lu H-Z, Zhang S-B, Shen S-Q (2015) High-field magnetoconductivity of topological semimetals with short-range potential. *Phys Rev B* 92.